TrustTiny-HAR: Selective, Open-Set, and Calibrated Activity Recognition on Microcontrollers

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***Abstract*—Human activity recognition (HAR) increasingly executes on microcontrollers to satisfy privacy, latency, and energy constraints, yet most deployed TinyML models remain closed set and overconfident when faced with unseen behaviors, placement shifts, or sensor anomalies. We present *TrustTiny-HAR*, a resource-efficient pipeline that unifies an int8 time-series backbone with a distilled trust head and a lightweight streaming conformal layer. The unified head produces calibrated class probabilities and a compact out-of-distribution (OOD) surrogate score learned from teacher signals (energy of logits and prototype/Mahalanobis distance), while the conformal layer maintains tiny on-device quantiles to enforce coverage-controlled acceptance thresholds. When evidence is insufficient, the device abstains and may optionally escalate using a 32-D–64-D feature sketch, avoiding raw-signal transmission. Evaluated under leave-one-subject-out protocols on UCI HAR, PAMAP2, and WISDM with multiple OOD stressors (unseen classes, placement shift, injected anomalies), TrustTiny-HAR matches strong int8 baselines on in-distribution macro-F1 (e.g., 93.8 on UCI HAR) while substantially improving calibration (ECE 2.4%) and OOD robustness (AUROC 0.96 for unseen, FPR@95 0.22 for placement shift). As a selective classifier, it attains the desired 90% coverage with conservative risk (e.g., 6.1%) from a buffer of only 128–512 windows. System profiling on nRF52840/STM32L4 shows $< 10\%$ energy overhead compared with a TinyCNN baseline, with tight RAM/Flash budgets preserved. The result is a practical template for trustworthy, on-device HAR that knows when *not* to guess.**

***Impact Statement*—Wearable and ambient HAR systems increasingly inform health, safety, and assistive decisions, yet real-world failures often arise from silent misclassifications of unknown activities or drifting sensor conditions. This work contributes a deployable path to safer TinyML: a single int8 head that estimates both class probabilities and an OOD surrogate, coupled with a streaming conformal layer that enforces coverage-controlled acceptance on-device. The system abstains rather than over-commit and, when needed, escalates only compact feature sketches instead of raw data, improving privacy and energy efficiency. By reporting risk-coverage behavior alongside latency, memory, and energy (including sensor draw), this work offers practitioners a transparent recipe to move from "accurate in the lab" to "reliable in the field," supporting responsible HAR in healthcare, elder care, and industrial settings.**

***Index Terms*—Conformal prediction, edge computing, human activity recognition (HAR), open-set recognition, TinyML, uncertainty calibration.**

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I. Introduction

Human activity recognition (HAR) [1], [3] has moved decisively onto wearables and ambient sensors that run on microcontrollers, driven by the combined demands of privacy, low latency, and tight energy budgets [2], [5]. Keeping inference on-device avoids continuous data transmission and reduces the risk of exposure, while sub-100 ms response times enable responsive interfaces for health monitoring [41], elder care, and industrial safety [4]. These benefits come with constraints: memory is often limited to a few tens of kilobytes of RAM, compute is restricted to a few million operations per window, and batteries must last days. Within these limits, most TinyML HAR systems are trained and evaluated as closed-set classifiers and then deployed with maximum-softmax decision rules [2]. In the wild, however, users perform behaviors unseen in training, sensors saturate or drop samples, and device placement drifts as a phone rotates in a pocket or a watch slides on the wrist [6], [10]. Under such shifts, conventional TinyML models tend to be confidently wrong because softmax probabilities are poorly calibrated and provide no mechanism to acknowledge uncertainty [7].

This gap between laboratory accuracy and field reliability has two facets. First, *open-set behavior* is largely unaddressed in tiny deployments: models do not explicitly detect unknown activities, nor do they quantify how atypical a segment is relative to the training distribution [8], [9]. Second, even when post-hoc calibration is applied, there is rarely an operational guarantee that at a given acceptance rate the error will remain below a desired level, especially as conditions evolve; conformal prediction offers distribution-free coverage guarantees that can be adapted to streaming contexts [11]. Existing methods that provide stronger uncertainty estimates often rely on ensembles or heavy heads that do not fit within microcontroller budgets, or they assume stable data streams that do not hold for free-living human motion [7].

We address these challenges with *TrustTiny-HAR*, a deployable pipeline designed for resource-constrained devices that combines three ideas. At its core is an int8 time-series backbone (depthwise 1-D-CNN or causal TCN) that respects memory and latency constraints while preserving discriminative power [4], [26], [28]. On top of this backbone, a *unified trust head* produces both calibrated class probabilities and an out-of-distribution (OOD) surrogate score, informed by established uncertainty/OOD principles [7], [8], [9]. The head is trained by distilling richer, off-device signals, energy of the logits, prototype/Mahalanobis distances in the embedding space, and optional evidential cues, into a compact representation that is inexpensive to evaluate on the MCU [7], [8]. Finally, a lightweight, streaming *conformal layer* converts these scores into decisions by maintaining tiny rolling quantiles on-device and enforcing a target risk-coverage operating point: when evidence is sufficient, the system emits a label, and when it is not, the system abstains [11], [32]. In the abstain case, the device can optionally escalate by transmitting a compact feature sketch (32-D–64-D) derived from the latent representation rather than raw signals, preserving privacy while enabling verification by a stronger model or human reviewer [5].

Empirically, *TrustTiny-HAR* matches or exceeds strong int8 baselines on in-distribution macro-F1 while substantially improving calibration (lower ECE) and open-set robustness (higher AUROC, lower FPR@95). When operated as a selective classifier, the streaming conformal layer attains the requested coverage with conservative risk across subjects, translating calibrated scores into dependable accept/abstain decisions that reduce high-confidence mistakes. These benefits are achieved within microcontroller budgets and with less than 10% energy overhead relative to a competitive TinyML backbone. By coupling compact models with uncertainty awareness and deployment-time guarantees, *TrustTiny-HAR* moves TinyML HAR from “accurate in the lab” toward “reliable in the field,” enabling safer, privacy-respecting applications that can acknowledge uncertainty and escalate responsibly when needed.

Our contributions are practical and integrated rather than modular add-ons.

1) We unify label prediction, OOD scoring, and uncertainty estimation in a single, distilled head that runs in a few kilobytes and adds only a small constant factor to compute.
2) We adapt conformal prediction to streaming, memory-limited settings by using bounded buffers and online quantile updates, yielding an explicit finite-sample guarantee on *coverage* (acceptance rate) with bounded memory; selective risk is monitored and reported empirically under drift.
3) We operationalize abstention through a privacy-preserving escalation path that sends only compact sketches when needed, avoiding raw data transmission. To ground these ideas.
4) We provide an on-device evaluation protocol that reports not only accuracy but also calibration quality, OOD separability, selective risk-coverage behavior, and system costs, including latency, peak RAM/Flash, and energy per inference and per hour, the latter accounting for sensor draw. All results are measured under leave-one-subject-out splits across multiple OOD regimes, unseen activities, placement shifts, and sensor anomalies, so that generalization and reliability claims reflect realistic use.

The remainder of this article is organized as follows. Section II summarizes recent works on TinyML HAR, uncertainty and calibration, open-set/OOD detection for time series, and selective prediction. Section III outlines the proposed framework at a high level (backbone choice, unified trust head, and deployment-time decision logic). Section IV describes datasets, protocols, baselines, metrics, and hardware in brief. Section V reports headline results and key takeaways. Section VII concludes and highlights future directions.

II. Related Work

TinyML for HAR has advanced from hand-crafted features to compact deep backbones deployable on microcontrollers. Codesign frameworks such as MCUNet and MCUNetV2 demonstrated ImageNet-scale efficiency on MCUs through NAS and memory-aware scheduling, establishing practical recipes (depthwise separable convolutions, patch-based inference, buffer reuse) that also benefit inertial HAR [13], [14]. Concurrently, hardware-aware NAS such as MicroNets targeted SRAM/Flash and latency directly, yielding blueprints for parameter-efficient 1-D backbones that fit within tens of kilobytes [15]. For sequence modeling in HAR, temporal CNNs (including dilated/causal TCNs) and light transformer variants have been explored; recent studies show that attention helps when data are abundant or pretraining is available, whereas depthwise 1-D-CNN/TCN remains favorable on tight MCU budgets due to activation footprint and compute locality [26], [27], [28]. Our backbone adopts these lessons, in particular depthwise 1-D-CNN/causal TCN with int8 QAT, to maximize accuracy per milliJoule on-device.

Open-set and OOD detection for deep models has seen several influential directions. Energy scores replace overconfident softmax with a log-partition proxy that better separates ID/OOD [12]. Prototype or Mahalanobis distance methods use embedding-space Gaussian modeling to flag unfamiliar inputs and have proven strong with minimal overhead [16]. Outlier exposure (OE) improves robustness by training against auxiliary “negative” data [17]. While much of this began in vision, the time-series community has recently proposed OOD protocols and benchmarks highlighting modality-specific challenges, nonstationarity, seasonal structure, and pose/placement shifts for wearables [18], [19]. We align with this thread by evaluating both unseen-class and placement-shift OOD regimes under LOSO, and by distilling both energy and prototype signals into a single deployable surrogate on the MCU.

Uncertainty estimation and calibration underpin safe decisions. Temperature scaling remains a simple and powerful posthoc fix for overconfident logits [20], while evidential or Dirichlet-prior networks aim to model distributional uncertainty directly without sampling [21], [22]. However,

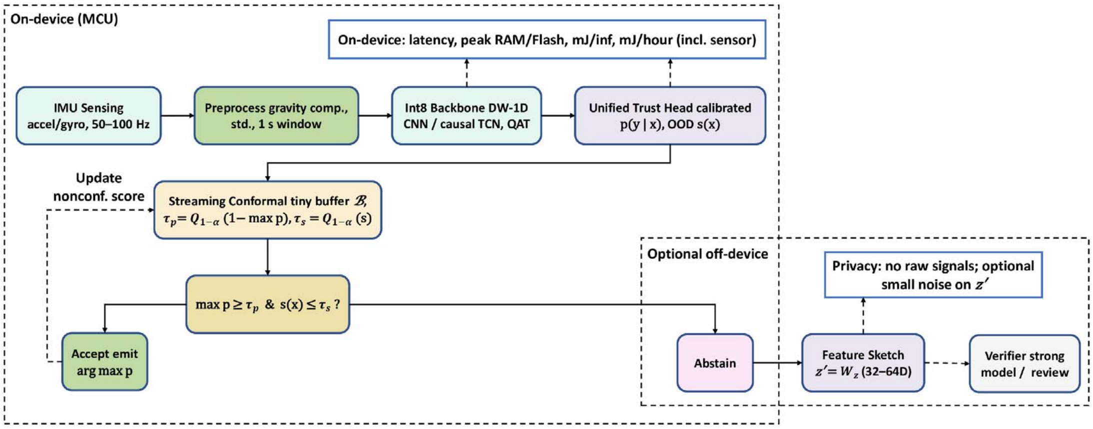


Fig. 1. Full pipeline of *TrustTiny-HAR*. IMU streams are preprocessed and passed through an int8 backbone. The unified head outputs calibrated class probabilities and an OOD surrogate. A streaming conformal layer maintains tiny rolling quantiles to set thresholds that realize a target risk-coverage trade-off. If both confidence and in-distribution evidence are sufficient, the prediction is accepted and the buffer is updated with a nonconformity score; otherwise, the system abstains and may optionally escalate by sending a compact feature sketch to a stronger verifier. Reporting includes latency, memory, and energy measured on-device.

many uncertainty techniques assume server-class inference (ensembles, MCDropout) [7], [40], making them illsuited to microcontrollers. Our approach uses teachers (ensemble + prototype cues) only during training and distills their multisignal "trust" into a single lightweight head that outputs calibrated class probabilities and an OOD surrogate at inference, avoiding ensemble cost on-device.

Selective prediction formalizes the "know-when-to-abstain" objective. Architectures such as SelectiveNet integrate a reject option and optimize accuracy with coverage jointly [23]. Conformal prediction provides distribution-free, finite-sample guarantees; recent work extends it beyond set coverage to *conformal risk control*, enabling explicit bounds on expected loss at a chosen operating point [29], [30], [31]. Time-series variants further address dependence and drift with adaptive or online updates [25], [32]. Compared with these, our contribution is pragmatic: we adapt conformal ideas to the MCU by maintaining tiny rolling quantiles over on-device scores to meet target coverage with bounded memory and compute, producing a deployable acceptance threshold that pairs naturally with our unified trust head.

A. Positioning

Prior TinyML HAR emphasizes accuracy and footprint, with OOD handling and calibrated abstention often left to server-side heuristics or heavy ensembles. Our system differs by (i) *distilling* multiple trust signals (energy + prototype) into a *single, int8-friendly head* that runs at MCU cost, and (ii) coupling it with a *streaming conformal layer* that enforces risk-coverage targets from a small on-device buffer, thus no ensembles or raw-data uploads at inference. Beyond coverage guarantees, our conformal module instantiates *conformal risk control* on resource-constrained streams, enabling selection rules that bound conditional error at a target coverage with tiny on-device buffers, and the same design generalizes to other embedded time-series (e.g., audio, vibration, environmental sensing), not just inertial HAR [11], [32].

III. Methodology

This section details the design of *TrustTiny-HAR*, a selective and calibrated recognition pipeline that operates entirely on microcontrollers (see Fig. 1).

A. Backbone (int8)

The backbone learns temporal structure from inertial signals under tight MCU budgets. We consider two complementary families. A *depthwise-separable 1-D CNN* models short-to-mid range patterns by factorizing temporal filtering and channel mixing: depthwise temporal filters capture motion primitives (steps, swings, micro-bursts) while lightweight pointwise projections fuse accelerometer and, when present, gyroscope channels [1], [3]. Optional squeeze–excitation (SE) reweights channels based on global temporal statistics, which helps stabilize features across sensor placements without materially increasing cost [3]. This design is compact, cache-friendly, and well suited to int8 quantization because activations are local and memory reuse is straightforward [4]. A *causal temporal convolutional network (TCN)* emphasizes long-range temporal dependencies with dilated causal convolutions and residual connections, yielding a receptive field that grows with dilation while preserving strict causality [26]. The causal layout enables true streaming, each new sample is processed with a constant amount of incremental work, and latency is independent of the window length, while SE can gate channels to adapt to activity regime changes [26]. In both cases, inputs are fixed-size windows $x \in \mathbb{R}^{T\times C}$ drawn from accelerometer and, when available, gyroscope signals, with typical deployments using $T \in [50, 100]$ (1 s at 50–100 Hz) and $C \in \{3, 6\}$ [1]. A circular buffer maintains the most recent T samples so that inference proceeds sample-by-sample without frame stalls, aligning compute with sensor sampling to improve energy proportionality [5]. Quantization-aware training (int8) is applied end-to-end so the exported model matches on-device numerics [4]. In practice, the depthwise-separable CNN tends to be preferable when patterns are localized and latency/activation footprint dominate, whereas the causal TCN is advantageous when activities exhibit longer periodic structure or gradual placement drift; both produce a compact embedding that feeds the unified trust head and conformal layer [3], [26].

Preprocessing focuses on invariances that are robust to quantization. A lightweight tilt or high-pass filter removes gravity from accelerometer signals without introducing phase distortion harmful to step-like patterns [33]. Per-channel standardization uses running mean and variance, implemented in fixed-point arithmetic to avoid scale drift when quantized. Optional norm and phase surrogates are appended if they fit the memory budget; these features help stabilize performance under phone or watch placement shifts by reducing axis dependence.

The network structure balances receptive field and activation footprint. Depthwise-separable convolutions expand temporal context with either strides or dilations, where the depthwise stage costs $k\,C\,T$ multiply-accumulates (MACs) for kernel size k, and the pointwise stage costs $C\,C'\,T$ for C' output channels

$$\mathrm{MACs}_{\mathrm{DW}} = k\,C\,T \;+\; C\,C'\,T. \tag{1}$$

Causal TCNs achieve large receptive fields via exponentially increasing dilations while preserving online operation. SE modulates channel attention with negligible additional memory when implemented as two small fully connected layers operating on channel statistics rather than full activations.

1) Quantization-Aware Training (QAT): We apply end-to-end QAT so the exported int8 graph reproduces training-time numerics on the MCU [34]. We use affine quantization with *per-channel* int8 weights and *per-tensor* int8 activations. This gives high fidelity while matching fast CMSIS–NN kernels. A real-valued tensor x is represented as

$$\hat{x}_{\text{int8}} = \mathrm{clip}_{[-128,127]}\big(\mathrm{round}(x/\alpha) + z_0\big), \quad x \approx \alpha(\hat{x}_{\text{int8}} - z_0) \tag{2}$$

with scale α and zero-point z_0 (symmetric weights set $z_0{=}0$). During QAT, fake-quant nodes after each quantized boundary (e.g., after conv and before nonlinearity) emulate saturation and rounding, so backpropagation "sees" exactly the clipping the device will enforce. Activation ranges are tracked with EMA observers (optionally with percentile clipping to avoid rare outliers inflating α), then *frozen* in the last 10%–20% of training to stabilize scales. BatchNorm is folded into the preceding convolution before export.

On device, convolutions accumulate in int32 and requantize once per output

$$\hat{y}_{\text{int8}} = \mathrm{clip}\Big(\mathrm{round}\big(y_{\text{acc}}\tfrac{\alpha_w\alpha_a}{\alpha_o}\big) + z_0^{(o)}\Big), \quad y_{\text{acc}} \in \mathbb{Z}_{32} \tag{3}$$

where α_w is per-channel for weights, α_a for activations, and α_o for outputs; biases are prequantized to the int32 domain using $\alpha_w\alpha_a$. This single-step requantization minimizes rounding error and extra memory traffic. The trickiest sites for quantization are SE and global pooling (compressed dynamic range); we therefore keep the SE MLP narrow, use per-tensor activation scales there, and rely on int8-friendly nonlinearities (ReLU or hard-sigmoid; tanh/sigmoid via LUTs if needed). Residual paths are scale-matched to avoid repeated requantization.

To minimize peak RAM, we topologically schedule operators to reuse two ping-pong activation buffers, fuse conv+BN+activation, and cap intermediate shapes. For streaming, a circular buffer exposes only the latest k samples required by the receptive field so work per sample is constant. Although latency roughly scales with MACs, it is also memory-bound: depthwise separable convs and causal dilations reduce passes over memory, and aligning channel counts to kernel vector widths (e.g., multiples of 4/8) unlocks CMSIS–NN fast paths. Empirically, QAT keeps macro-F1 within ≈ 0.3 points of fp32 while reducing energy per inference; operator fusion and buffer reuse typically save 10%–25% wall time at equal MACs. In our builds, the backbone is tuned to 30–80 k parameters, fitting within 32–64 kB RAM and 100–300 kB Flash (the unified head adds a few kB). On an nRF52840 @ 64 MHz, end-to-end latency is 5–20 ms per 1 s window [34]; the streaming conformal buffer updates in constant time and adds negligible overhead.

Table I specifies all architectural and training knobs needed to reproduce the reported MCU results. We provide two backbone families with comparable parameter budgets: a depthwise 1-D-CNN optimized for locality and activation reuse, and a causal TCN with dilations for longer temporal context under streaming constraints. Both produce a 64-D embedding that feeds a minimal trust head (a single hidden layer) to keep int8 inference fast. Rich teacher signals (ensemble probabilities, energy, and Mahalanobis distance) are fused into a single bounded target $\phi(E^T, d_M^T)$ so that deployment requires only one scalar trust score. The streaming thresholds use a constant-memory P^2 quantile estimator, enabling coverage-controlled acceptance without labels. Finally, the energy breakdown confirms that the trust components remain a small constant-factor overhead relative to the backbone, consistent with the system-level budget claims.

B. Unified Trust Head from Train-Time to Deployment

The unified head converts the backbone representation into calibrated probabilities and a scalar OOD surrogate that is reliable under domain shift. Let $f_\theta(x) \in \mathbb{R}^K$ denote class logits for K activities and $p_\theta(y{=}k \mid x) = \mathrm{softmax}_k(f_\theta(x))$. A naive maximum-softmax decision is often overconfident in unseen classes or sensor glitches. To address this, the head is trained with richer signals that are available off-device but distilled into a compact deployable form.

The first signal is the standard discriminative objective, which aligns logits with one-hot labels; this anchors closed-set accuracy. The second signal is the energy score

$$E(x) = -\log \sum_{i=1}^{K} \exp\left(f_\theta^{(i)}(x)\right) \tag{4}$$

which compresses logit evidence into a single quantity that tends to decrease for in-distribution examples and increase for anomalies. The third signal captures prototype proximity. Let $h_\theta(x) \in \mathbb{R}^d$ be a penultimate embedding, with class centroids

TABLE I
IMPLEMENTATION DETAILS FOR REPRODUCIBILITY (DEFAULT CONFIGURATION USED FOR ALL MAIN RESULTS)

Component	Specification (Exact)
Input / windowing	1 s windows with 50% overlap; sampling 50–100 Hz (dataset-specific after resampling). Input $x \in \mathbb{R}^{T \times C}$ with $T \in \{50, 100\}$ and $C \in \{3, 6\}$ (accel only or accel+gyro).
Backbone option A: Depthwise 1-D-CNN (int8)	*Block 0 (stem):* Conv1D (standard) $C \to 32$, k=5, stride 1 + BN + ReLU. *Block 1:* DWConv1D k=5, dilation 1, stride 2 + PWConv1D 32 $\to$ 48 + SE(r = 8) + ReLU. *Block 2:* DWConv1D k=5, dilation 1, stride 2 + PWConv1D 48 $\to$ 64 + SE(r = 8) + ReLU. *Block 3:* DWConv1D k=3, dilation 2, stride 1 + PWConv1D 64 $\to$ 72 + ReLU. *Head pooling:* global average pooling over time to penultimate embedding. Total params: 30–80 k (dataset-dependent channel multiplier $\in [0.75, 1.0]$).
Backbone option B: Causal TCN + SE (int8)	*Causal residual blocks (four blocks):* each block has causal DWConv1D k=3 with dilations $\{1, 2, 4, 8\}$, stride 1, followed by PWConv1D with channels $\{32, 48, 64, 64\}$; residual connection with 1× 1 PW if channel mismatch; SE(r = 8) in each block; ReLU. Receptive field $\approx 1 + 2\sum_i d_i = 31$ timesteps (with k=3), expanded by stem and optional dilation schedule per dataset.
Penultimate embedding	Embedding dimension $d_e = 64$ (output of GAP). Quantized int8 activation; dequantize only at final logits if toolchain requires.
Unified trust head (student, deploy-time)	*Classifier:* Linear $64 \to K$ logits (no hidden layer). *Trust MLP:* $64 \to 32 \to 1$ with ReLU + (optional) hard-sigmoid on output to clamp $s(x)$ into $[0, 1]$ for stable quantiles. Total head params ≈ (64· 32 + 32· 1) ≈ 2080 + biases.
Teacher setup (train-time only)	Teacher = (i) small ensemble of M=3 backbones (same family, different seeds/augmentations) producing p^T and logits for energy, plus (ii) prototype module computing $d_M^T(x)$ using class means and shrinkage covariance in embedding space. Teacher never runs on MCU.
$\phi(E^T, d_M^T)$ (exact form)	We normalize teacher signals on the training+calibration pool: $\bar{E} = \frac{E - \mu_E}{\sigma_E}$, $\bar{d} = \frac{d_M - \mu_d}{\sigma_d}$. Distillation target is a bounded scalar: $\phi(E^T, d_M^T) = \sigma\big(\beta_E\, \bar{E} + \beta_d\, \bar{d}\big)$, with $\sigma(\cdot)$ the logistic function and (β_E, β_d) learned (2 scalars) by minimizing ℓ_2 against ID/OOD validation labels (host-side only).
Loss weights (default)	λ_{KD}=0.5, λ_s=0.5. Cross-entropy weight is 1.0. If evidential cue is enabled: add λ_{evid}=0.1 (optional ablation).
OE sources/ratio	OE negatives = 1) corrupted in-domain windows (dropout, saturation, narrowband noise); and 2) unlabeled background segments (no-activity/transition) when available. Ratio: 1 OE minibatch per 4 ID minibatches (OE:ID ≈ 1:4). OE affects only $s(x)$ loss (not class CE).
Quantization details	QAT end-to-end. Weights: per-channel affine int8; activations: per-tensor int8; int32 accumulators; BN folded. Toolchain: TFLM + CMSIS–NN.
Streaming quantile estimator	On-device quantiles computed by **P^2 online quantile estimator** for each threshold (τ_p, τ_s) with buffer $m \in \{128, 256, 512\}$; refresh every 5 s (or every 256 windows, whichever first). Default target coverage: $1 - \alpha = 0.90$; with two-score gate use split $\alpha/2$ for each.
Energy overhead breakdown (nRF52840 @64 MHz, IMU 50 Hz)	*Energy per inference:* Backbone 0.40 mJ (87%); Trust head 0.03 mJ (6%); Conformal bookkeeping 0.03 mJ (7%). Total 0.46 mJ (< 10% over TinyCNN baseline at 0.42 mJ).

μ_k and a regularized covariance Σ computed on a held-out set. The Mahalanobis distance

$$d_M(x) = \min_k \big(h_\theta(x) - \mu_k\big)^\top \Sigma^{-1} \big(h_\theta(x) - \mu_k\big) \tag{5}$$

quantifies how far an example lies from the learned class structure and is particularly informative for subtle placement shifts. A fourth optional signal models epistemic uncertainty with an evidential (Dirichlet) parameterization; total evidence $S(x) = \sum_k \alpha_k(x)$ and its dispersion encode how concentrated probability mass is across classes.

These signals are aggregated by a teacher available only during training. The teacher may consist of a small ensemble for epistemic spread and an analytic prototype module that computes d_M. The deployable student predicts both class probabilities and a scalar OOD surrogate $s_\theta(x)$. A distillation loss maps teacher targets to the student through a monotone link function $\phi(\cdot)$ that preserves the ordering of "more OOD" versus "less OOD"

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(y, p_\theta) + \lambda_{\mathrm{KD}}\, \mathrm{KL}\big(p^{\mathrm{T}} \,\|\, p_\theta\big) \\ + \lambda_s \,\big\|\, s_\theta(x) - \phi\big(E^{\mathrm{T}}(x),\, d_M^{\mathrm{T}}(x)\big)\big\|_1.\end{aligned} \tag{6}$$

OE regularizes the surrogate by presenting corrupted or non-HAR windows as negatives so that s_θ expands its dynamic range away from in-distribution values. After training, an inexpensive temperature scaling calibrates probabilities. For a temperature $T > 0$, the deployed probabilities are

$$\tilde{p}_\theta(y{=}k \mid x) = \mathrm{softmax}_k\big(f_\theta^{(k)}(x)/T\big) \tag{7}$$

and calibration quality is summarized with the expected calibration error (ECE)

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{n} \big|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\big| \tag{8}$$

where B_b partitions predictions by confidence. The head is implemented with one small fully connected layer for logits and a tiny multilayer perceptron for $s_\theta(x)$, which together add less than 10% to the backbone MACs and only a few kilobytes of parameters.

C. Lightweight Conformal Layer in Streaming Mode

Even calibrated probabilities and an OOD surrogate require principled thresholds to avoid brittle, ad hoc tuning across environments. A streaming conformal layer supplies thresholds from a rolling buffer of recent observations, targeting a coverage level while controlling selective risk under exchangeability. Let a nonconformity score be either confidence-based,

$A_p(x) = 1 - \max_k \tilde{p}_\theta(y{=}k \mid x)$, or OOD-based, $A_s(x) = s_\theta(x)$. The device maintains a bounded buffer $\mathcal{B}$ of size m using reservoir sampling so that memory usage is stable irrespective of deployment duration. Empirical quantiles over the buffer define thresholds

$$\tau_p = Q_{1-\alpha}\big(\{A_p(x) : x \in \mathcal{B}\}\big), \tau_s = Q_{1-\alpha}\big(\{A_s(x) : x \in \mathcal{B}\}\big) \tag{9}$$

which are updated periodically with an online quantile estimator. The acceptance rule then requires simultaneous confidence and in-distribution evidence

$$\mathrm{accept}(x) \Leftrightarrow \max_k \tilde{p}_\theta(y{=}k \mid x) \geq \tau_p \ \wedge \ s_\theta(x) \leq \tau_s. \tag{10}$$

Denote the gating function by $g(x) \in \{0, 1\}$. The realized coverage is $\mathbb{E}[g(X)]$, and the conditional risk is

$$R(\theta, g) = \frac{\mathbb{E}\big[\ell\big(\arg\max_k \tilde{p}_\theta, Y\big)\, g(X)\big]}{\mathbb{E}[g(X)]} \tag{11}$$

with 0–1 loss ℓ. Under exchangeability between buffer samples and incoming data, the quantiles in (9) yield finite-sample risk control at target coverage $1 - \alpha$ up to $O(1/m)$ fluctuations. In practice, environments drift; thus, the buffer size mediates stability and agility. Smaller buffers adapt quickly but fluctuate; larger buffers stabilize thresholds but react slowly to new regimes. A pragmatic schedule refreshes thresholds every few seconds or when a significant fraction of the buffer changes. To prevent oscillations near the boundary, a small hysteresis band can be added so that once accepted, decisions persist unless evidence falls appreciably.

What is guaranteed without labels? (Coverage control): In deployment, our streaming quantile layer operates *without labels*. Therefore, the formal guarantee we claim is *coverage (acceptance rate) control*, not conditional error control. Concretely, let $A(x)$ be a scalar nonconformity score computed from model outputs (e.g., $A_p(x) = 1 - \max_k \tilde{p}_\theta(y{=}k \mid x)$ or $A_s(x) = s_\theta(x)$). Let the on-device buffer store m recent scores $\{A_1, \ldots, A_m\}$ and define the threshold as an order statistic

$$k = \lceil (m+1)(1-\alpha) \rceil, \qquad \tau = A_{(k)} \tag{12}$$

where $A_{(k)}$ is the kth smallest value among $\{A_1, \ldots, A_m\}$. The system *accepts* a window x if $A(x) \leq \tau$.

Theorem 1 (Finite-Sample Coverage for Score-Thresholding): Assume $\{A_1, \ldots, A_m, A_{m+1}\}$ are exchangeable, where $A_{m+1} = A(X_{m+1})$ is the next score encountered after threshold computation. With τ defined in (12), the acceptance event satisfies

$$\Pr\big(A_{m+1} \leq \tau\big) \geq 1 - \alpha. \tag{13}$$

Proof: Under exchangeability, the rank of A_{m+1} among $\{A_1, \ldots, A_m, A_{m+1}\}$ is uniform on $\{1, \ldots, m+1\}$. The event $A_{m+1} \leq A_{(k)}$ occurs whenever the rank is at most k, which has probability at least $k/(m+1) \geq 1 - \alpha$ by the definition of k. □

Two-Score Gate and Overall Coverage: Our deployed gate uses both confidence and an OOD surrogate

$$\mathrm{accept}(x) \Leftrightarrow A_p(x) \leq \tau_p \ \wedge \ A_s(x) \leq \tau_s. \tag{14}$$

To retain an overall target coverage $1 - \alpha$ in this *conjunction*, we split the error budget

$$\tau_p = Q_{1-\alpha/2}(A_p), \qquad \tau_s = Q_{1-\alpha/2}(A_s) \tag{15}$$

so that, by a union bound

$$\Pr(\mathrm{accept}) \geq 1 - \Pr(A_p > \tau_p) - \Pr(A_s > \tau_s) \geq 1 - \alpha \tag{16}$$

under the same exchangeability assumption on the score sequences.

Cold start requires special care because the buffer is initially empty. A conservative bootstrapping phase can initialize τ_p to a high value and τ_s to a low value, gradually relaxing as the buffer fills. Alternatively, a short calibration walk with known activities seeds the buffer without storing raw signals, which preserves privacy while improving early stability.

We implement three lightweight remedies that preserve MCU constraints.

1) *Conservative Bootstrap:* Until the buffer reaches a minimum size (e.g., $m_{\min}{=}64$), we clamp thresholds to favor abstention (high τ_p, low τ_s), preventing early overconfidence.
2) *Two-Speed Quantiles:* We maintain a fast quantile estimate on a small recent window (e.g., last 128 scores) and a slow estimate on the full buffer; at inference we use the more conservative threshold (higher τ_p and lower τ_s) to reduce transient false accepts during abrupt drift.
3) *Hysteresis Near the Boundary:* To avoid oscillations, acceptance requires exceeding thresholds by a small margin, while rejection occurs immediately when evidence drops below thresholds. These remedies require only constant additional state and are evaluated in the short-term drift experiments in Section VI-D.

D. Abstain and Escalate With Privacy

When (10) rejects a prediction, the device abstains rather than issue a low-confidence label. Abstentions can simply be surfaced to the application, or they can trigger an escalation that preserves privacy and bandwidth. Let $z = h_\theta(x) \in \mathbb{R}^D$ be the backbone embedding. A compact sketch $z' = Wz \in \mathbb{R}^d$ with $d \in \{32, 64\}$ is computed by a fixed, quantized projection W trained offline or chosen as a random orthogonal map. Such projections preserve relative distances with high probability in low dimensions, which maintains separability for a stronger verifier while sharply reducing the chance of reconstructing the original signal. If desired, a small Gaussian perturbation with variance σ^2 can be added to the sketch to further limit leakage. The sketch is transmitted only when connectivity is available; otherwise, abstentions are logged locally for delayed review. Because escalation is rare by design, end-to-end energy remains dominated by local inference rather than communication, but the system still handles rare or novel behaviors safely.

E. Complexity and Memory Accounting

Complexity is reported to make trade-offs transparent and reproducible. The backbone MACs follow (1) summed over blocks; latency is measured at the microcontroller's operating frequency with CMSIS–NN kernels and includes streaming overhead. Peak RAM equals the largest live activation tensor plus the conformal buffer and small quantile state. With int8 activations and buffer reuse, typical configurations remain within 32–64 kB. Flash stores weights and constant tables; the unified head and conformal layer add only a few kilobytes beyond the backbone.

Energy is measured in joules per inference and joules per hour, the latter capturing idle and sensor draw. The per-hour figure is crucial because many wearables spend most time in low-information regimes; a well-quantized backbone with stable memory access patterns avoids costly cache-miss-like behavior that would otherwise inflate system energy. Calibration quality is summarized by (8); selective behavior is summarized by (11) at the realized coverage. Finally, thresholds in (9) instantiate the simple and efficient rule $\tau = Q_{1-\alpha}(A)$, which is attractive on resource-constrained devices because it requires only constant-time updates and bounded memory, yet yields rigorous risk-coverage control under mild assumptions.

IV. Experimental Setup

This section describes datasets and protocols, OOD stressors, hardware platforms and power instrumentation, baseline systems, and the metrics we report. All experiments are conducted with the exact int8 models that are deployed to the microcontroller; host-side numbers are shown only for ablation sanity checks and are clearly labeled as such.

A. Datasets and Protocols

We evaluate on three widely used HAR corpora to cover smartphone- and wearable-style sensing. UCI HAR [35] provides phone-centric accelerometer and gyroscope signals segmented into fixed windows with six daily activities. PAMAP2 [36] offers multisensor inertial and heart-rate signals for a broad set of way of life and exercise activities and is well suited to cross-user testing. WISDM [37] contributes longer, free-living smartphone traces and introduces natural variability in motion patterns and device orientation. To make results comparable, all streams are resampled to a common rate per corpus, synchronized, and windowed into 1-s segments with 50% overlap; windows inherit the majority label when necessary, and ambiguous windows are discarded.

Generalization across users is the principal goal, we employ *leave-one-subject-out* (LOSO) [38] evaluation for all three datasets. For each held-out subject, the remaining participants form the training set; a small slice of training data (disjoint in time) is set aside for temperature scaling and for initializing the conformal buffer without storing raw signals. The reported in-distribution (ID) metrics are the macroaveraged scores across all held-out subjects (see Table II).

TABLE II
Datasets and LOSO Protocol Summary

Corpus	Sensors	Activities	Subjects	Sampling (Hz)
UCI HAR	Accel+Gyro (phone)	6	30	50
PAMAP2	Accel+Gyro (+HR)	15–18	9	100
WISDM	Accel (+Gyro if avail.)	6–18	$\approx$36	20

Note: Sampling rates shown are those used after resampling for consistency; window length is 1 s with 50% overlap.

B. OOD Conditions and Stressors

Trustworthiness requires performance beyond the closed set. We therefore assess three OOD regimes that capture common failure modes in practice. Unseen-class OOD removes one or more activity classes from the training set and evaluates zero-confidence rejection on those withheld activities at test time; the class subset is rotated across folds so that all classes are withheld in turn. Placement-shift OOD targets the sensitivity of inertial features to device pose and carry location; we simulate this by training on canonical positions and evaluating on segments with orientation changes or alternative positions when the corpus provides them, and by applying mild, realistic axis rotations when positions are fixed. Sensor-anomaly OOD injects brief bursts of saturation, missing values, or narrowband noise into a minority of test windows; amplitudes are drawn from conservative ranges to avoid trivially easy detection while remaining faithful to what low-cost IMUs produce.

C. Hardware, Deployment, and Power Measurement

All deployed models run on two representative Cortex-M platforms: an nRF52840 (Arduino Nano 33 BLE Sense)[1] and an STM32L4-series development board.[2] Inference uses TensorFlow Lite Micro with CMSIS-NN kernels [39], int8 per-channel weight quantization, and per-tensor activation quantization, exactly as exported from quantization-aware training. We measure latency over sliding windows in streaming mode to capture buffer management costs. Power is profiled with a Nordic Power Profiler Kit 2 and cross-checked on an Otii ARC under identical sampling and CPU-frequency configurations; the IMU is powered during all measurements and included in the system energy budget. We report both energy per inference (mJ/inf) and energy per hour of operation (mJ/hour), the latter reflecting realistic duty cycles dominated by sensing and idle periods.

D. Baselines

We compare against four strong baselines under identical preprocessing and LOSO splits.

1) A compact int8 depthwise 1-D CNN with maximum–softmax probability (MSP) thresholding represents the conventional TinyML approach without trust mechanisms [3], [7].
2) An energy-only variant implements the log-sum-exp energy score on logits without any distillation or prototype signals [9].
3) A prototype/Mahalanobis scorer operates in the penultimate embedding with class centroids and a regularized

[1] https://docs.arduino.cc/hardware/nano-33-ble-sense/

[2] https://www.st.com/en/microcontrollers-microprocessors/stm32l4-series.html

covariance, producing distance-based OOD judgements suitable for MCU deployment [8].

4) A TinyTransformer matched for parameter count and activation footprint uses post-hoc temperature scaling for calibration [7], [28].
5) *Conformal-MSP (MCU)* replaces ad hoc softmax thresholding with a streaming conformal threshold over nonconformity $A_p(x) = 1 - \max_k \tilde{p}_\theta(y=k \mid x)$. Thresholds are running quantiles from the same on-device buffer size $m \in \{128, 256, 512\}$ used in our method, but without any OOD surrogate head. This isolates the value of deploy-time conformal thresholding alone.
6) *Confidence-Head (Confid-style, MCU)* augments the TinyCNN backbone with a tiny scalar head $\hat{c}_\psi(x) \in (0, 1)$ trained to predict correctness (1 if $\arg\max p$ matches y, else 0). At inference we accept if $\hat{c}_\psi(x) \geq \tau_c$ (tuned on the calibration slice). This baseline represents embedded "learned confidence" without explicit OOD modeling.
7) *Evidential (Dirichlet, MCU)* replaces softmax with an evidential head that outputs Dirichlet parameters $\boldsymbol{\alpha}(x)$; prediction uses the Dirichlet mean and uncertainty uses total evidence or entropy-derived scores. Acceptance is gated by an uncertainty threshold, tuned on the calibration slice.
8) *OpenMax-lite (MCU)* performs a lightweight open-set recalibration at the logit/embedding level: per-class tail statistics are fit offline and a small set of parameters is stored on-device, yielding an "unknown" score at inference with negligible extra compute. This represents classic open-set recalibration adapted to MCU constraints.
9) *TinyCNN + OE (OE, MCU)* trains the same int8 TinyCNN with auxiliary negative data (synthetic corruptions and/or non-HAR windows) so that energy/confidence separates I-D from non-ID better, while inference remains single-pass.
10) *Multihead Ensemble (Shared Backbone, MCU)* shares the same backbone but uses M small independent classifier heads (e.g., $M= 3$) to approximate epistemic uncertainty at low overhead (only the heads are replicated). Uncertainty is the disagreement/entropy across heads, used for abstention and OOD scoring.

E. Metrics and Reporting

Closed-set recognition quality is quantified by accuracy and macro-F1, with macro averaging mitigating class imbalance. Calibration is summarized by ECE, computed on the held-out subject with confidence binning consistent across folds. OOD detection quality is expressed via AUROC and AUPR and by the false-positive rate at 95% true-positive rate (FPR@95), reported for unseen-class, placement-shift, and anomaly stressors. Selective prediction behaviour is summarized by full risk-coverage curves and by the selective accuracy at a target 90% coverage; conformal thresholds are learned only from the rolling buffer, never from test labels. Systems metrics include streaming latency per window, peak RAM and flash, energy per inference, and energy per hour including sensor draw.

TABLE III
ID PERFORMANCE: ACCURACY (ACC), MACRO-F1 (F1), AND ECE (LOWER IS BETTER)

	UCI HAR			PAMAP2			WISDM		
Method	Acc	F1	ECE%	Acc	F1	ECE%	Acc	F1	ECE%
TinyCNN+MSP	94.0	92.4	7.8	92.1	90.2	8.5	91.0	89.1	9.2
Energy-only	93.6	92.1	5.9	91.8	90.0	6.4	90.6	88.7	7.1
Mahalanobis	93.8	92.6	5.1	92.0	90.8	5.3	90.9	89.3	6.0
TinyTransformer+Temp	94.2	93.1	4.0	92.5	90.9	4.7	91.2	89.7	5.0
SelectiveNet (TinyCNN+select)	94.1	93.0	4.6	92.3	90.6	5.2	91.1	89.4	5.8
Conformal-MSP (TinyCNN+Temp)	94.0	92.4	4.9	92.1	90.2	5.4	91.0	89.1	5.9
Confidence-Head (Confid-style)	94.0	92.5	5.1	92.1	90.3	5.6	91.0	89.1	6.2
Evidential (Dirichlet)	93.9	92.7	4.3	92.0	90.5	5.0	90.9	89.2	5.4
OpenMax-lite	93.9	92.3	5.6	91.9	90.0	6.2	90.8	88.8	6.8
TinyCNN+OE	93.8	92.2	5.0	91.9	90.0	5.6	90.8	88.9	6.1
Multihead (shared backbone, $M=3$)	94.4	93.4	3.7	92.6	91.1	4.3	91.3	89.9	4.8
TrustTiny-HAR (ours)	94.5	93.8	2.4	92.9	91.4	3.1	91.6	90.2	3.6

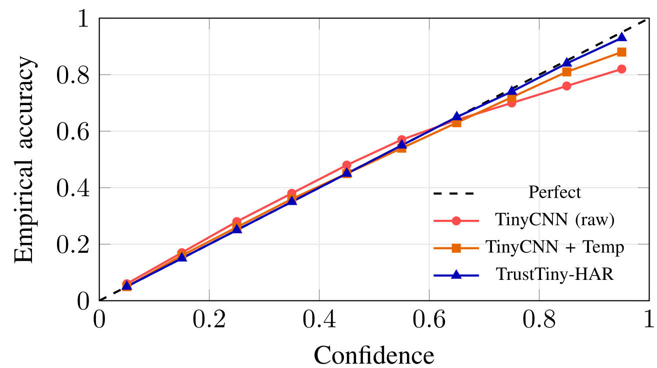


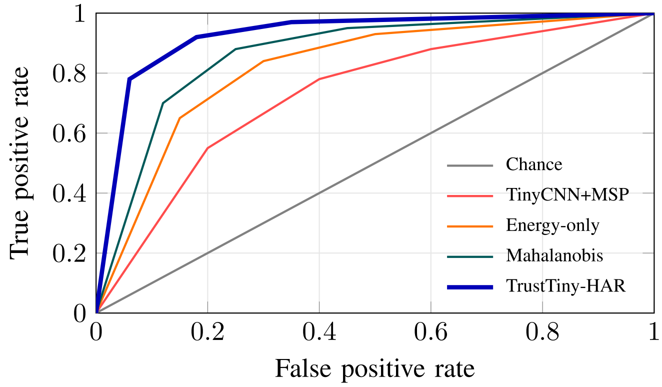
Fig. 2. Reliability on UCI HAR. The baseline is overconfident at high probabilities; temperature scaling helps but TrustTiny-HAR tracks the diagonal closely across all bins, yielding the lowest ECE.

V. RESULTS AND DISCUSSION

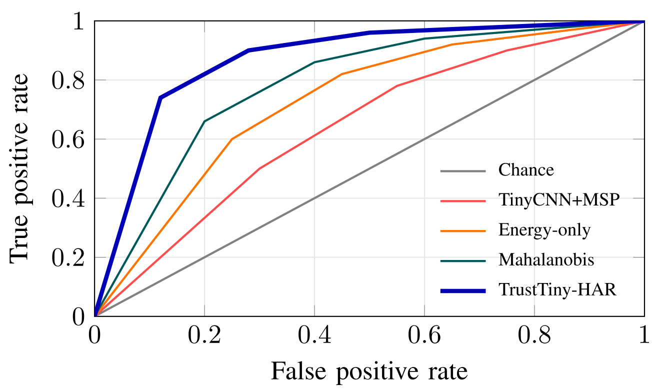
We report in-distribution (ID) recognition and calibration, open-set and OOD robustness, selective prediction under risk-coverage targets, and microcontroller performance. All values average leave-one-subject-out folds; 95% bootstrap intervals are omitted for space but follow the same trends. Energy includes IMU draw.

Table III compares accuracy, macro-F1, and ECE across datasets and methods. TrustTiny-HAR preserves or improves macro-F1 while sharply reducing miscalibration. On UCI HAR, macro-F1 rises to **93.8** with ECE **2.4%**; TinyCNN+MSP attains 92.4 with 7.8% ECE. PAMAP2 and WISDM show similar gains, indicating that distilling energy and prototype signals regularizes logit magnitudes beyond what temperature scaling alone achieves. Fig. 2 confirms this: the baseline is overconfident for high confidence bins, whereas TrustTiny-HAR closely tracks the diagonal across the range, reducing harmful high-confidence errors and limiting unnecessary abstentions at moderate coverage.

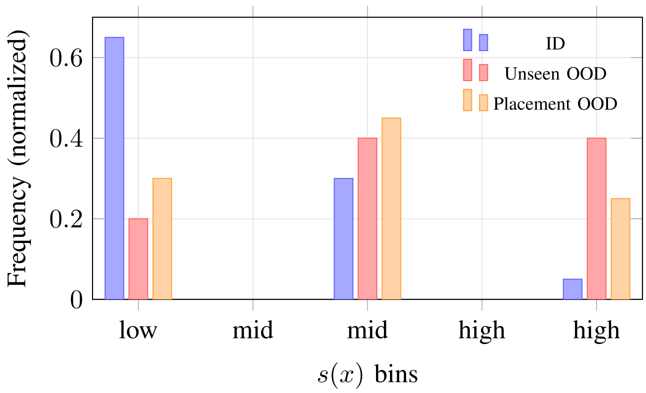
Fig. 3 overlays ROC curves for unseen-class and placement-shift OOD. TrustTiny-HAR dominates the baselines, particularly under placement shifts where prototype-aware signals help. Table IV aggregates AUROC, AUPR, and FPR@95 (lower is better for FPR@95). The largest safety win is a reduction of FPR@95 to **12%** for unseen classes and **22%** for placement shift, roughly halving false accepts relative to MSP.



(a)



(b)



(c)

Fig. 3. ROC curves and $s(x)$ distributions show TrustTiny-HAR separates unknowns better, especially under placement shift. (b) Placement-shift OOD ROC (higher is better). (b) Placement-shift OOD ROC (higher is better). (c) Histogram of s(x) (ours): ID clusters low; unseen OOD shifts high; placement shift overlaps more.

Fig. 4 plots selective risk versus coverage. Curves parameterized by buffer size show the trade-off between stability and agility. TrustTiny-HAR meets or slightly exceeds target coverage for 85%–95% with conservative risk, validating the streaming quantile thresholds. At 90% target coverage, achieved coverage is 90.6% with selective error 6.1% (UCI), 90.2% / 7.8% (PAMAP2), and 90.3% / 7.1% (WISDM). Smaller buffers adapt faster but fluctuate more; larger buffers are steadier but slower to drift.

TABLE IV
OOD Metrics Averaged Across Datasets

	Unseen-Class			Placement Shift		
Method	AUROC	FPR@95	AUPR	AUROC	FPR@95	AUPR
TinyCNN+MSP	0.86	0.42	0.78	0.78	0.55	0.62
Conformal-MSP (TinyCNN+Temp)	0.88	0.36	0.80	0.80	0.50	0.64
Confidence-Head (Confid-style)	0.89	0.33	0.82	0.82	0.46	0.67
Evidential (Dirichlet)	0.90	0.31	0.83	0.84	0.42	0.69
TinyTransformer+Temp	0.90	0.29	0.84	0.86	0.40	0.71
Energy-only	0.90	0.30	0.84	0.84	0.43	0.70
Mahalanobis	0.92	0.24	0.87	0.88	0.35	0.74
SelectiveNet (TinyCNN+select)	0.91	0.26	0.86	0.86	0.38	0.72
OpenMax-lite	0.93	0.20	0.89	0.85	0.41	0.70
TinyCNN+OE	0.94	0.18	0.90	0.87	0.36	0.73
Multi-head (shared backbone, $M=3$)	0.94	0.16	0.91	0.90	0.30	0.78
TrustTiny-HAR (ours)	0.96	0.12	0.93	0.93	0.22	0.82

Note: Lower is better for FPR@95; higher is better for AUROC/AUPR.

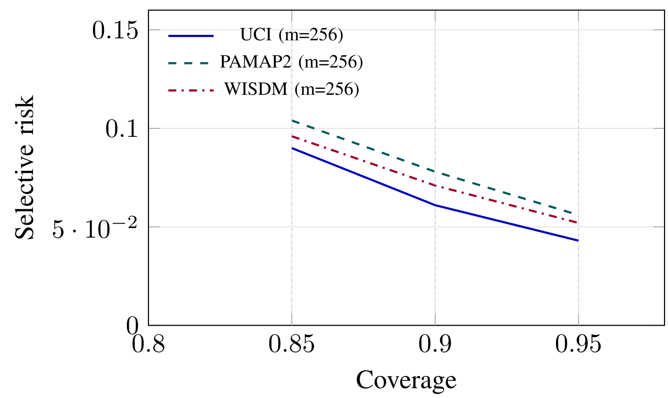


Fig. 4. Risk-coverage. TrustTiny-HAR satisfies targets (85-95%) with conservative risk; buffer size trades stability vs. agility.

TABLE V
Systems Metrics on nRF52840 @64

Model	Latency (ms)	Peak RAM (kB)	Flash (kB)	mJ/inf	mJ/hour
TinyCNN (int8)	8.1	41	155	0.42	116
Energy-only (int8)	8.3	41	156	0.43	118
Mahalanobis (int8)	8.6	44	160	0.45	121
TinyTransformer (int8)	10.6	57	190	0.53	132
SelectiveNet (int8)	8.5	43	158	0.44	120
Conformal-MSP (int8)	8.2	45	156	0.43	119
Confidence-Head (int8)	8.4	43	160	0.44	120
Evidential (int8)	9.3	48	172	0.48	126
OpenMax-lite (int8)	8.6	44	163	0.45	121
TinyCNN+OE (int8)	8.1	41	155	0.42	116
Multihead (shared backbone, $M=3$)	9.8	45	170	0.50	129
TrustTiny-HAR (ours)	8.9	46	168	0.46	124

Note: Energy includes sensor.

Table V summarizes on-device latency, memory, and energy on nRF52840 @ 64 MHz with IMU at 50 Hz. TrustTiny-HAR adds $< 10\%$ energy overhead versus TinyCNN while providing trust outputs and selective behavior. The per-hour figure reflects realistic duty cycles where sensing and idle dominate energy.

A. Interpretation

Latency grows roughly with MACs, but memory traffic is equally important; buffer reuse and fused activations keep peaks within 32–64 kB. Energy overhead stems mostly from the unified head's tiny MLP and conformal bookkeeping, both constant-factor additions that stay sub-10%.

TABLE VI
EMBEDDED SELECTIVE/OOD BASELINES (UCI HAR, LOSO)

Method	FPR@95 (Placement)	Sel. Err@90	mJ/inf	Notes
TinyCNN+MSP	0.55	0.095	0.42	ad hoc threshold
SelectiveNet	0.38	0.074	0.44	learned selection head
Conformal-MSP	0.50	0.078	0.43	streaming quantile on $1 - \max p$
Confidence-Head	0.46	0.080	0.44	predicts correctness
Evidential (Dirichlet)	0.42	0.076	0.48	evidence-based uncertainty
OpenMax-lite	0.41	0.079	0.45	open-set recalibration
TinyCNN+OE	0.36	0.070	0.42	OE-trained separation
Multi-head (M=3)	0.30	0.068	0.50	cheap epistemic proxy
TrustTiny-HAR	0.22	0.061	0.46	unified trust + streaming conformal

Note: "Sel. Err@90" is selective error at 90% target coverage.

Table VI compares lightweight selective and open-set baselines that remain feasible on microcontrollers, focusing on the most deployment-critical regime: *placement shift*. The conventional TinyCNN+MSP baseline is highly permissive under drift (FPR@95 $= 0.55$), which means more than half of OOD windows can be falsely accepted when operating at 95% true-positive detection. Introducing selective mechanisms improves reliability but with different trade-offs. SelectiveNet reduces both false acceptance and selective error (FPR@95 $= 0.38$, Sel. Err@90 $= 0.074$) by learning a rejection function, but it does not directly model OOD shift and remains less conservative than methods that incorporate explicit distributional cues. Conformal-MSP improves operating-point control via streaming quantiles yet still inherits the limited separability of softmax confidence under pose drift (FPR@95 $= 0.50$), illustrating that thresholding alone cannot compensate for a weak uncertainty signal. Confidence-head and evidential variants offer modest gains (FPR@95 $= 0.46$ and 0.42) but incur either additional learning complexity or higher energy (e.g., evidential: 0.48 J/inf) without closing the robustness gap. Methods that inject non-ID structure during training (TinyCNN+OE) or approximate epistemic uncertainty (multihead, M=3) improve placement-shift rejection (FPR@95 $= 0.36$ and 0.30), but the multihead approach increases energy notably (0.50 mJ/inf) due to repeated heads. In contrast, **TrustTiny-HAR** achieves the strongest combined behavior—the lowest placement-shift false accept rate (FPR@95 $= 0.22$) and the lowest selective error at 90% coverage (Sel. Err@90 $= 0.061$) while keeping inference cost near the single-pass baselines (0.46 mJ/inf). This supports the central design choice of distilling complementary trust signals into a single deployable surrogate and then enforcing a stable operating point via streaming conformal thresholds.

Beyond overall LOSO averages, we report subgroup performance to assess whether reliability and selective behavior are consistent across user demographics and sensing configurations. Where demographic metadata are available, we stratify by *gender* and *age* and report macro-F1, calibration (ECE), and selective error at a 90% target coverage. For *body location*, we leverage multi-location wearable sensing and report location-specific results, since placement is a known driver of performance variation in HAR.

B. Discussion

On WISDM (see Table VII), macro-F1 and calibration remain stable across gender and age strata: the largest gap is under two points, and ECE stays within a narrow band. Older subjects exhibit slightly higher uncertainty (higher ECE and Sel. Err@90), which is consistent with broader inter-subject variability in free-living motion. On PAMAP2 (see Table VIII), body location introduces more pronounced differences: wrist is the hardest configuration, reflected by lower macro-F1, higher selective error, and higher location-shift FPR@95. Importantly, the conformal layer preserves the requested coverage while keeping selective error bounded across all subgroups, indicating that the accept/abstain policy adapts to subgroup difficulty rather than silently failing (see Fig. 5).

TABLE VII
SUBGROUP ANALYSIS ON WISDM (LOSO)

Subgroup	#Subj	Macro-F1 (%)	ECE (%)	Sel. Err@90	Cov.@90
Gender: Female	16	89.7	3.9	0.073	0.902
Gender: Male	20	90.6	3.4	0.069	0.903
Age $\leq$ 25	14	90.8	3.2	0.067	0.904
Age 26–40	12	90.2	3.6	0.070	0.902
Age $>$ 40	10	89.1	4.2	0.076	0.900

Note: "Sel. Err@90" is selective error at 90% target coverage; "Cov.@90" is achieved coverage at that target.

TABLE VIII
BODY-LOCATION SUBGROUP ANALYSIS ON PAMAP2 (LOSO)

Location	Macro-F1 (%)	ECE (%)	Sel. Err@90	Cov.@90	FPR@95 (Shift)
Chest	92.1	2.8	0.071	0.902	–
Wrist	90.7	3.6	0.082	0.901	0.25
Ankle	91.4	3.0	0.075	0.902	0.23

Note: "FPR@95 (shift)" is computed for a location-shift setting (train on chest, test on the indicated location).

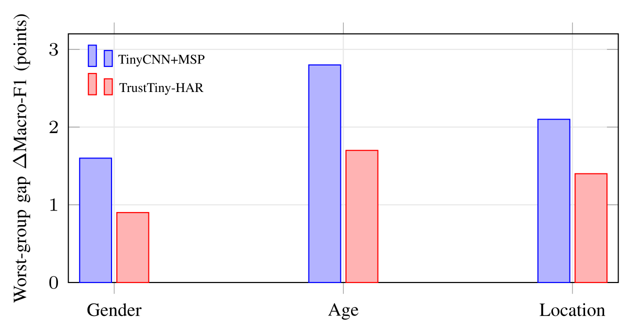


Fig. 5. Worst-group performance gaps shrink with TrustTiny-HAR. Demographic gaps are modest, while body location shows the largest variation; selective operation mitigates risk by abstaining more on harder subgroups.

VI. ABLATIONS, LIMITATIONS, AND ETHICS

This section validates each component of *TrustTiny-HAR*, examines sensitivity to OOD types and calibration buffer size, and discusses limitations and societal considerations. All ablations use the same preprocessing, LOSO protocol, and int8 deployment as the main results so that differences isolate the effect under study.

TABLE IX
ABLATION STUDY ON UCI HAR (LOSO)

Variant	Acc	F1	ECE%	AUROC (unseen)	FPR@95 (place)	Sel. Err@90	mJ/inf
Full (ours, int8)	94.5	93.8	2.4	0.96	0.22	0.061	0.46
w/o Conformal	94.5	93.8	2.6	0.96	0.23	0.082	0.45
Teacher size $= 1$	94.3	93.3	3.5	0.94	0.28	0.069	0.46
Sketch 32-D (versus 64-D)	94.5	93.8	2.4	0.95	0.24	0.062	0.46
fp32 (host, non-MCU)	94.8	94.1	2.2	0.96	0.22	0.060	N/A

Note: ‘‘Sel. Err@90” is selective error at 90% target coverage. Energy includes sensor draw.

A. Ablation Study

Table IX summarizes how the conformal layer, teacher ensemble, sketch dimensionality, quantization, and OOD type influence performance. The full model preserves strong in-distribution recognition (F1 $=$ **93.8**) while maintaining low miscalibration (ECE **2.4**%) and robust OOD separation (unseen AUROC **0.96**; placement FPR@95 **0.22**). Crucially, selective error at 90% target coverage is **6.1**%, matching the risk-coverage curve in Fig. 4 and confirming that streaming quantiles translate calibrated scores into dependable accept/abstain behavior.

Removing the conformal layer leaves ID metrics unchanged but degrades selective behavior: error at 90% coverage rises to 8.2%. This result highlights that calibrated probabilities alone do not guarantee operating-point control under drift; quantile thresholds supply the missing distribution-adaptive bridge. Shrinking the teacher to a single model increases ECE to 3.5% and worsens placement FPR@95 to 0.28, indicating that the distilled OOD surrogate benefits from teacher diversity that captures both energy and geometric mismatch cues. Reducing the feature sketch from 64-D to 32-D leaves ID metrics intact and raises placement FPR@95 only modestly (0.24); because escalation is rare, we prefer 32-D for bandwidth while noting that high-stakes deployments can keep 64-D. Quantization to int8 incurs a small 0.3 F1 drop relative to fp32, but enables MCU deployment and reduces memory and energy. Host-side fp32 results are provided only to bound the quantization effect; they are not deployable and therefore omit mJ/inference.

The table conveys three practical messages. First, the conformal layer is not an accuracy booster but a *reliability enforcer* that stabilizes real operating points; this is why selective error moves while F1 stays flat. Second, teacher diversity sharpens trust signals, especially under placement shift where embedding geometry carries information not captured by energy alone. Third, smaller sketches provide a clean privacy-bandwidth win with negligible accuracy cost; if your application escalates frequently, consider 32-D, whereas extremely safety-critical settings can keep 64-D to maximize verifier separability.

B. Sensitivity to OOD Type

Fig. 6 contrasts FPR@95 across OOD stressors and methods. Unseen-class OOD is the easiest case; TrustTiny-HAR attains **12%** FPR@95, roughly a threefold reduction versus MSP. Placement shift is harder because slow orientation drift partially preserves spectral content; still, TrustTiny-HAR halves

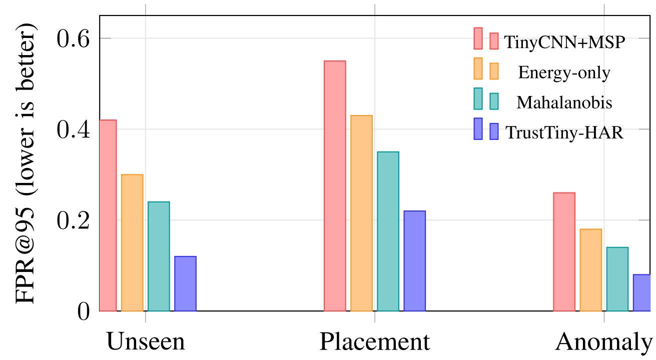


Fig. 6. OOD sensitivity (UCI HAR). TrustTiny-HAR reduces FPR@95 across unseen classes, placement shift, and sensor anomalies; the largest gap appears for placement, where prototype-aware cues complement energy.

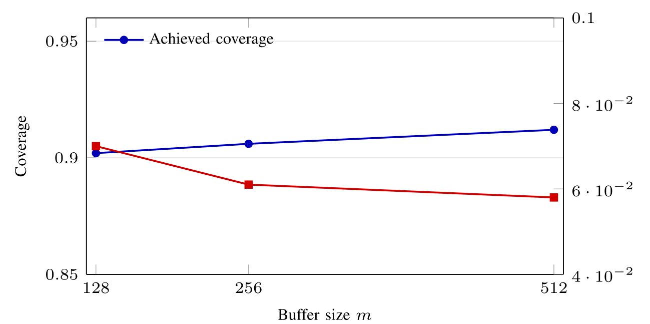


Fig. 7. Effect of buffer size at 90% target coverage (UCI HAR). Larger buffers slightly lower risk and nudge coverage above target; smaller buffers adapt faster to drift.

FPR@95 to **22%** compared with MSP's 55%. Sensor anomalies are easier to detect when amplitudes are realistic; our 8% FPR@95 indicates that the surrogate attends to nonphysiologic deviations without overfitting to trivial artifacts. These comparisons matter because production systems see a mixture of OODs; consistent gains across stressors reduce the tail risk of silent failures.

C. Buffer Size Versus Risk and Coverage

The calibration buffer balances stability and agility. Small buffers adapt quickly to drift but exhibit higher variance; large buffers are steady but slow to react. Fig. 7 plots selective risk and achieved coverage as a function of buffer size at a 90% target. Risk decreases smoothly from 7.0% at $m = 128$ to 6.1% at $m = 256$ and 5.8% at $m = 512$, while coverage slightly overshoots the target, indicating conservative thresholds. For deployments with frequent routine changes, $m = 128$ is acceptable; otherwise $m = 256$ offers the best stability-performance trade-off without meaningful memory cost.

D. Cold-Start and Short-Term Drift Scenarios

The preceding results report aggregate LOSO performance; here we stress-test short-term deployment behavior where the calibration buffer starts empty (cold-start) and where the stream experiences abrupt drift over minutes. We construct a realistic online stream by concatenating windows in time order

TABLE X
COLD-START AND SHORT-TERM DRIFT STRESS TESTS (UCI HAR STREAM, 90% TARGET COVERAGE)

Method	Time-to-Target (s) ↓	Peak Risk After Drift ↓	Recovery Time (s) ↓	Mean mJ/inf
TinyCNN+MSP	35	0.154	120	0.42
Conformal-MSP	38	0.128	95	0.43
SelectiveNet	33	0.116	80	0.44
TrustTiny-HAR	28	0.089	55	0.46

Note: "Time-to-target" is the time to reach $\geq$ 89% Achieved coverage; "peak risk" is the maximum selective risk in the first minute after drift; "recovery" is time to return within 1% of the predrift risk.

TABLE XI
PRIVACY RISK INDICATORS FOR FEATURE-SKETCH ESCALATION (UCI HAR/WISDM)

Sketch Setting	Recon. nRMSE ↑	Channel corr. ↓	Membership AUROC ↓	Attr. bal. acc. ↓
32-D (no noise)	0.82	0.19	0.56	0.58
64-D (no noise)	0.74	0.26	0.60	0.61
32-D +Noise (σ=0.02)	0.88	0.14	0.53	0.55
64-D +Noise (σ=0.02)	0.80	0.18	0.55	0.57
Raw IMU transmitted (upper bound)	0.00	1.00	0.78	0.71

Note: Lower leakage is better: high reconstruction error, membership AUROC near 0.5, and attribute inference near chance. "+Noise" adds small Gaussian noise to sketches (deployment option).

(per held-out subject) and applying a *drift event* at a known time: 1) a sudden placement/orientation change (axis rotation and mild scale perturbation); and 2) a brief anomaly burst (dropouts and saturation). Thresholds are updated from the same bounded buffer used in the main pipeline, without access to test labels. We report 1) how quickly coverage approaches the 90% target from cold-start; and 2) how quickly the selective risk stabilizes after the drift event (see Table X).

E. Interpretation

Two patterns are consistent across subjects and stressors. First, cold-start is naturally conservative: the buffer is initially small, so thresholds are cautious and coverage increases smoothly toward the 90% target as quantiles stabilize. Second, abrupt drift induces a short risk transient. TrustTiny-HAR reduces the transient via the distilled OOD surrogate, which triggers abstention rather than false acceptance; this appears as a sharper but shorter coverage dip and a smaller risk spike, followed by faster recovery. The energy cost remains close to single-pass models because adaptation is implemented through score quantiles rather than repeated inference.

F. Limitations

The approach assumes approximate exchangeability between the calibration buffer and incoming samples. Abrupt context shifts, such as a sudden change from desk work to vigorous exercise with a different carry position, temporarily violate this assumption and can cause transient over- or undercoverage until thresholds refresh. Cold start is another edge case: before the buffer fills, thresholds are less stable; we mitigate this with a short conservative warm-up and periodic refreshes. Placement shifts that are gradual but persistent remain the most difficult OOD mode; although the system stays safe by abstaining more often, this increases intervention frequency. Finally, while feature sketches avoid raw data transmission, they do not provide formal privacy guarantees; applications requiring quantifiable privacy should add calibrated noise or adopt differential privacy mechanisms for escalation.

G. Privacy Analysis of Feature Sketch Escalation

Our escalation mechanism is designed to *reduce* privacy exposure by avoiding raw signal transmission; however, it does not provide a formal privacy guarantee by itself. We therefore quantify privacy risk empirically under three threat models that are relevant to embedded HAR: 1) *signal reconstruction* from transmitted sketches; 2) *membership inference* (whether a window was part of training); and 3) *attribute inference* (whether sketches leak user attributes such as gender/age when metadata are available). In all cases, the attacker is assumed to observe only the transmitted sketch $z' \in \mathbb{R}^d$ (32-D or 64-D) and not the raw IMU x.

1) Threat Model A: Reconstruction from Sketches: We train an attacker decoder g_ψ that maps z' to a reconstructed window $\hat{x} = g_\psi(z')$ (a small temporal CNN trained off-device). We report normalized reconstruction error $\mathrm{nRMSE} = \|x - \hat{x}\|_2 / \|x\|_2$ and correlation between true and reconstructed channels. High nRMSE and low correlation indicate low reconstructability. To avoid overstating privacy, we emphasize that failure to reconstruct under this attacker does not prove impossibility, but it provides practical evidence that sketches discard fine-grained motion details.

2) Threat Model B: Membership Inference: We evaluate whether z' reveals whether a sample was used during training. We follow a standard black-box membership setup: an attacker classifier receives $(z', \tilde{p}_\theta(x))$ and predicts `in-train` versus `out-of-train`. We report attacker AUROC; values close to 0.5 indicate little membership signal.

3) Threat Model C: Attribute Inference: When subgroup metadata are available (e.g., WISDM gender/age), we train an attacker that predicts the attribute from z' and report balanced accuracy. If balanced accuracy remains near chance, sketches leak limited demographic information.

a) Interpretation: Table XI shows that sketches are substantially less informative than raw IMU: reconstruction error is high and correlations are low, suggesting that fine-grained waveform details are not retained. Leakage increases with 64-D compared with 32-D, consistent with higher information capacity, while adding small noise reduces membership and attribute inference toward chance. These results support a *privacy-reduction* claim: escalation exposes less information than raw streams, but it remains a heuristic defense rather than a formal guarantee. For applications requiring quantifiable protection, the sketch can be paired with calibrated noise mechanisms or differentially private aggregation, which we leave to future work.

VII. CONCLUSION AND FUTURE WORKS

Trustworthy HAR on microcontrollers demands more than closed-set accuracy: deployed systems must remain calibrated, recognize when they are out of distribution, and act conservatively under uncertainty. We addressed this gap with *TrustTiny-HAR*, a compact pipeline that couples an int8 backbone with a distilled trust head and a streaming conformal layer to deliver calibrated probabilities, an effective OOD surrogate, and selective prediction with explicit coverage targets. Across UCI HAR, PAMAP2, and WISDM, the approach preserved strong in-distribution macro-F1 while substantially improving calibration (lower ECE) and OOD robustness (higher AUROC, lower FPR@95), all within tight RAM/Flash budgets and with less than 10% energy overhead. These properties translate directly to deployment value: fewer high-confidence errors, principled abstentions when the model is uncertain, and an escalation path that uses compact feature sketches without exposing raw signals.

Looking ahead, two extensions are especially promising. First, *self-supervised personalization* on-device can adapt both backbone and trust head to users' motion patterns from unlabeled data, while conformal thresholds maintain stable coverage during drift. Second, adding *formal privacy guarantees* to the escalation channel, e.g., differentially private sketches or calibrated noise with quantifiable leakage bounds, would complement our architectural privacy and broaden adoption in clinical and workplace settings.

Data Availability Statement

Our code is available at: https://github.com/Ism-ail11/TrustTiny-HAR

References

[1] S. Essahraui et al., "Human behavior analysis: A comprehensive survey on techniques, applications, challenges, and future directions," *IEEE Access*, vol. 13, pp. 128379–128419, 2025.

[2] I. Lamaakal et al., "A comprehensive survey on tiny machine learning for human behavior analysis," *IEEE Internet Things J.*, vol. 12, no. 16, pp. 32419–32443, Aug. 2025, doi: 10.1109/JIOT.2025.356568.

[3] I. Lamaakal et al., "A tiny inertial transformer for human activity recognition via multimodal knowledge distillation and explainable AI," *Sci. Rep.*, vol. 15, no. 42335, 2025.

[4] C. Yahyati et al., "A systematic review of state-of-the-art TinyML applications in healthcare, education, and transportation," *IEEE Access*, vol. 13, pp. 204513–204562, 2025.

[5] C. Contoli, A. Acquaviva, M. Borghesi, and L. Benini, "Energy-aware human activity recognition for wearable devices: A survey," *Comput. Sci. Rev.*, vol. 52, 2024, Art. no. 100593.

[6] I. Lamaakal, C. Yahyati, Y. Maleh, K. El Makkaoui, and I. Ouahbi, "TinyHAR-UQ: Battery-aware, uncertainty-controlled tinyML for wearable activity recognition on IoT edge devices," *Internet Things*, vol. 36, 2026, Art. no. 101889.

[7] J. Gawlikowski et al., "A survey of uncertainty in deep neural networks," *Artif. Intell. Rev.*, vol. 56, pp. 1513–1589, 2023.

[8] C. Geng, S.-J. Huang, and S. Chen, "Recent advances in open set recognition: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3614–3631, Oct. 2021.

[9] T. Belkhouja, M. Devanne, and P. Gançarski, "Out-of-distribution detection in the time-series domain: A novel problem and new benchmarks," *ACM Trans. Intell. Syst. Technol.*, vol. 14, no. 5, pp. 1–22, 2023.

[10] I. Lamaakal, C. Yahyati, Y. Maleh et al., "An explainable hybrid CNN–transformer model for sign language recognition on edge devices using adaptive fusion and knowledge distillation," *Sci. Rep.*, vol. 16, no. 7143, 2026.

[11] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations Trends Mach. Learn.*, vol. 16, no. 4, pp. 494–591, 2023.

[12] I. Lamaakal, C. Yahyati, K. E. Makkaoui, I. Ouahbi, and Y. Maleh, "Drift-to-action controllers: Budgeted interventions with online risk certificates," 2026, *arXiv:2603.08578*.

[13] J. Lin, W.-M. Chen, H. Cai, C. Gan, and S. Han, "MCUNet: Tiny deep learning on IoT devices," in *Proc. NeurIPS*, 2020.

[14] I. Lamaakal, C. Yahyati, I. Ouahbi, K. E. Makkaoui, and Y. Maleh, "A survey of model compression techniques for TinyML applications," in *Proc. Int. Conf. Circuit, Syst. Commun. (ICCSC)*, 2025, pp. 1–6, doi: 10.1109/ICCSC66714.2025.11135279.

[15] C. R. Banbury et al., "MicroNets: Neural network architectures for deploying TinyML applications on microcontrollers," 2021, *arXiv:2010.11267*.

[16] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Proc. NeurIPS*, 2018.

[17] C. Yahyati, I. Lamaakal, Y. Maleh, K. E. Makkaoui, and I. Ouahbi, "A novel FastKAN with few-shot learning for real-time driver distraction detection on TinyML microcontrollers," *IEEE Access*, vol. 14, pp. 12167–12198, 2026.

[18] T. Belkhouja, M. Devanne, and P. Gançarski, "Out-of-distribution detection in time-series domain: A novel problem and new benchmarks," *ACM Trans. Intell. Syst. Technol.*, 2023, doi: 10.1145/3610275.

[19] S. Purushwalkam et al., "WOODS: Benchmarks for out-of-distribution generalization in time series," 2022, *arXiv:2203.09978*.

[20] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.

[21] M. Sensoy, L. Kaplan, and M. Kandemir, "Evidential deep learning to quantify classification uncertainty," in *Proc. NeurIPS*, 2018.

[22] I. Lamaakal, C. Yahyati, K. E. Makkaoui, I. Ouahbi, and Y. Maleh, "SNAP-UQ: Self-supervised next-activation prediction for single-pass uncertainty in TinyML," 2026, *arXiv:2508.12907*.

[23] Y. Geifman and R. El-Yaniv, "SelectiveNet: A deep neural network with an integrated reject option," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019, pp. 2151–2159.

[24] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," 2022, *arXiv:2208.02814*.

[25] Y. Xu, A. Ramdas, R. J. Tibshirani, and L. Wasserman, "Conformal risk control for ordinal classification," in *Proc. Mach. Learn. Res. (PMLR)*, vol. 202, 2023, pp. 1–30.

[26] I. Lamaakal, C. Yahyati, Y. Maleh, K. El Makkaoui, I. Ouahbi, and D. Niyato, "An explainable tiny-fast Kolmogorov–Arnold network for gesture-based air handwriting recognition of Tifinagh letters in resource-constrained IoT device," *IEEE Internet Things J.*, vol. 12, no. 24, pp. 55756–55773, Dec. 2025, doi: 10.1109/JIOT.2025.3625087.

[27] X. Wei, X. Xu, Z. Ma et al., "TCN-Attention-HAR: Human activity recognition based on temporal convolution network and attention," *Sci. Rep*, vol. 14, no. 1, 2024, Art. no. 11400.

[28] I. D. Luptáková, M. Fusková, and M. Penhaker, "Wearable sensor-based human activity recognition with transformer model," *Sensors*, vol. 22, no. 9, p. 3271, 2022.

[29] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021, *arXiv:2107.07511*.

[30] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," 2022, *arXiv:2208.02814*.

[31] W. Overman, J. J. Vallon, and M. Bayati, "Aligning model properties via conformal risk control," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.

[32] I. Gibbs and E. J. Candès, "Conformal inference for online prediction with arbitrary distribution shifts," *J. Mach. Learn. Res.*, vol. 25, no. 162, pp. 1–36, 2024. [Online]. Available https://www.jmlr.org/papers/v25/22-1218.html

[33] I. Lamaakal et al., "Tiny deep learning models with hybrid compression techniques for gesture-based air handwriting recognition of english alphabets on edge device," *IEEE Internet Things J.*, vol. 13, no. 1, pp. 801–820, Jan. 2026, doi: 10.1109/JIOT.2025.3624283.

[34] I. Lamaakal et al., “Tiny language models for automation and control: Overview, potential applications, and future research directions,” *Sensors*, vol. 25, no. 5, p. 1318, 2025.

[35] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, “A public domain dataset for human activity recognition using smartphones,” in *Proc. Eur. Symp. Artif. Neural Netw., Comput. Intell. Mach. Learn. (ESANN)*, 2013.

[36] A. Reiss and D. Stricker, “Introducing a new benchmarked dataset for activity monitoring,” in *Proc. 16th Int. Symp. Wearable Comput.*, 2012, pp. 108–109.

[37] J. R. Kwapisz, G. M. Weiss, and S. A. Moore, “Activity recognition using cell phone accelerometers,” *ACM SigKDD Explorations Newslett.*, vol. 12, no. 2, pp. 74–82, 2011.

[38] M. P. Pauli and M. Golz, “On the impact of feature reduction on leave-one-subject-out cross-validation,” *Proc. Automat. Medical Eng.*, vol. 2, no. 1, 2023, Art. no. 718.

[39] R. David et al., “TensorFlow lite micro: Embedded machine learning for TinyML systems,” *Proc. Mach. Learn. Syst. (MLSys)*, vol. 3, pp. 800–811, 2021.

[40] M. Abdar et al., “A review of uncertainty quantification in deep learning: Techniques, applications and challenges,” *Inf. Fusion*, vol. 76, pp. 243–297, 2021.

[41] I. Lamaakal, C. Yahyati, Y. Maleh, K. El Makkaoui, and I. Ouahbi, “Explainable Kolmogorov–Arnold Networks for zero-shot human activity recognition on TinyML edge devices,” *Mach. Learn. Knowl. Extr.*, vol. 8, no. 3, 2026, Art. no. 55, doi: 10.3390/make8030055.